Online Resource 1

Expanded FDM-KLE baseline results for eight truth realizations

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This supplementary table reports the direct FDM-KLE baseline results for eight 3-mode truth realizations at $n_{\mathrm{obs}} = 200$ and observation seed 202. Truth labels are anonymized display labels used in the manuscript and do not encode the internal random seeds used to generate the fields. Cell values are reported as logK RMSE / hydraulic-head RMSE from closed-loop validation on the 201×201 grid. The weak-pass criterion is logK RMSE < 0.15 and head RMSE < 0.006; the strong-pass criterion is logK RMSE < 0.10 and head RMSE < 0.003.

Truth	$\sigma_{\log K}$	$\sigma = 0.005$	$\sigma = 0.01$	$\sigma = 0.02$
1	0.36	PASS (0.0203 / 7.3e-4)	PASS (0.0406 / 1.3e-3)	PASS (0.0815 / 2.5e-3)
2	0.71	PASS (0.0172 / 8.7e-4)	PASS (0.0344 / 1.4e-3)	PASS (0.0692 / 2.5e-3)
3	0.47	PASS (0.0184 / 6.9e-4)	PASS (0.0372 / 1.2e-3)	PASS (0.0750 / 2.3e-3)
4	0.40	PASS (0.0160 / 4.6e-4)	PASS (0.0321 / 8.7e-4)	PASS (0.0644 / 1.8e-3)
5	0.17	PASS (0.0200 / 5.4e-4)	PASS (0.0399 / 1.1e-3)	PASS (0.0800 / 2.2e-3)
6	0.34	PASS (0.0170 / 4.5e-4)	PASS (0.0342 / 9.1e-4)	PASS (0.0689 / 1.9e-3)
7	0.56	PASS (0.0175 / 4.7e-4)	PASS (0.0348 / 9.2e-4)	PASS (0.0696 / 1.9e-3)
8	0.80	PASS (0.0230 / 5.5e-4)	PASS (0.0450 / 1.1e-3)	PASS (0.0892 / 2.2e-3)

Median logK RMSE by noise level is 0.0180 at $\sigma = 0.005$, 0.0360 at $\sigma = 0.01$, and 0.0723 at $\sigma = 0.02$. All 24 cases satisfy both the weak- and strong-pass criteria.